

ECE 420 Project Proposal

Automated DJ System for Drum and Bass

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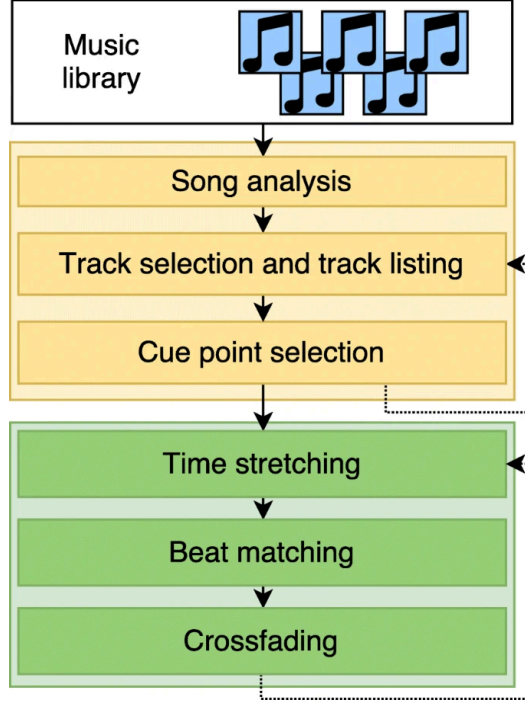
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I. INTRODUCTION

For our final project, we chose to build an automated DJ system for drum and bass. The goal is to create seamless music mixes using songs from a given playlist/library much like a human DJ does. DJing is a profession that requires skill, experience and expensive equipment. An automated DJ system such as this one will allow anyone with an interest in music an opportunity to save them a lot of trouble and lowers the barrier to enter creative music mixing. From casual listeners to clubs, cafes and radio stations, anyone can have the ability to generate custom mixes from their own music playlists. Apart from this, the technical implementation offers an intersection of artificial intelligence, signal processing and creativity. It's a great way to utilize machine learning, music information retrieval (MIR) such as tempo, beat and structure, and signal analysis to simulate human intuition.

As of now, we have a rough idea of how we want to apply the algorithm given in our chosen algorithm. The first step is to analyze the music. This would involve listening to each song and determining the beats, temp, downbeats, song structure, key and energy/style. In order to execute this, we plan on using MIR algorithms. The next step is to determine which songs to mix. In order for a seamless mix, the songs need to be chosen such that they match in key, have compatible styles and can transition naturally. Next is the cue points. This is about pinpointing the best spots to start or fade out songs. This would use the beat drops to make the transition feel more natural. Finally, there is the actual mixing which involves matching the beats using time stretching so that the tempos align, crossfading the songs and adjusting the drum and bass to make the blend smooth. This process is repeated across all the chosen songs to finally generate a set of songs that are fluidly mixed with perfect transitions and beat matching and essentially imitate the work of a human DJ.

II. OVERVIEW OF THE ALGORITHM



The system described in Vande Veire and De Bie’s paper replicates an automatic DJ framework using music information retrieval (MIR) techniques. The process this paper uses to produce seamless Drum and Bass mixes involves beat tracking, downbeat tracking, structural segmentation, track and cue point selection, and crossfading.

Beat tracking:

In this step, we estimate the tempo and phase of a track. The tempo is the beats-per-minute and the phase is the time difference between the first beat and the start of the signal. Given the insights that note onsets provide us related to beat tracking, such as from the repetition and loudness of onsets, our first step is to estimate the position of note onsets using an onset detection function.

$$\Gamma(m) = \sum_{k=1}^{40} HWR(X_{mel40}(m, k) - X_{mel40}(m - 1, k))$$

$X_{mel40}(m, k)$ is the energy of the frame m in the k th frequency bin, logarithmically spaced according to the Mel scale. $HWR()$ is the half-wave rectify operation - $HWR(x) = \max(x, 0)$. The beat period, in seconds, is estimated from the autocorrelation of the onset detection function.

$$A_{\Gamma}(l) = \frac{1}{M} \sum_m \Gamma_{HWR}(m) \cdot \Gamma_{HWR}(l + m)$$

From these autocorrelation results, we build a tempo detection curve $B(T)$ that measures how strongly the onsets repeat at a given lag time. The curve evaluates the autocorrelation at each potential lag and its multiples so lags that align with the true beat period produce stronger, more regular peaks (at their multiples).

$$\beta(\tau) = \frac{1}{N} \sum_n A(nL_\tau)$$

The beat period estimate is then chosen as the value of T that maximizes $B(T)$, corresponding to the most periodic and prominent pattern in the music.

$$\kappa = \operatorname{argmax}(\beta(\tau))$$

Once the tempo is known, the next step is to determine where the beats actually start within that period - the beat phase. To find the beat phase, the algorithm slides a one-beat-spaced sampling pattern across the onset detection function, summing the ODF values at every possible time shift ϕ between 0 and the estimated beat period. The phase that produces the largest sum is taken as the beat phase estimate.

$$\Phi(\phi) = \frac{1}{N} \sum_n \Gamma_{HWR}(nL_\kappa + L_\phi)$$

Downbeat Tracking:

Downbeat detection identifies the first beat of each measure, assuming a 4/4 time signature typical of Drum and Bass. First, the algorithm extracts per-beat “isolated” features such as loudness, Mel-band energy, and three ODFs (flux, complex spectral difference, and high-frequency coefficient).

$$x_{frame}^{[N_\kappa; N_\kappa]}(m) = x[N_\kappa(m): N_\kappa(m + 1)]$$

$$\chi_{loud}(m) = \text{loudness}(x_{frame}^{[N_\kappa; N_\kappa]}(m))$$

$$\chi_{mel}(m, k) = \chi_{mel12}(x_{frame}^{[N_\kappa; N_\kappa]}(m))(k)$$

$$\chi_{odfi}(m, k) = \Gamma_{HWR}^{(i)}(L_{beat}^{(m)} + k), 0 \leq k \leq L_\kappa, i \in \{flux, csd, hfc\}$$

However, we need to know how beats relate to one another to detect the rhythm. To do this, the system computes contextual features that describe how the current beat differs from the next 4 or 8 beats. These are used in a logistic regression classifier that estimates, for each beat, the probability of it being the 1st, 2nd, 3rd, or 4th beat in a measure. Then for each of the four possible options, the

highest of the corresponding summed log-probabilities of the beats lets us know which option is right and thus the proper sequence of downbeats.

Structural Segmentation:

The novelty-based structural segmentation method assumes that structural boundaries are marked by significant musical changes in audio. Knowing when these transitions occur is crucial for placing cue points or transitions when mixing. The algorithm uses structural similarity matrices (SSMs) which compare every moment in the track to every other moment, showing how similar certain features are. Changes in the SSM indicate a song's musicality has shifted. We use two SSMs: one based on MFCC features, capturing timbre, and another based on RMS energy, tracking loudness and dynamics.

$$S_{RMS}(m, n) = \left| \chi_{RMS}^{[N_k/2; N_k/4]}(m) - \chi_{RMS}^{[N_k/2; N_k/4]}(n) \right|$$

$$S_{MFCC}(m, n) = d_{cos}(\chi_{MFCC}^{[N_k/2; N_k/4]}(m) - \chi_{MFCC}^{[N_k/2; N_k/4]}(n))$$

Each SSM is analyzed with a novelty detection function, highlight points where similarity shifts sharply and thus music sections transition. The results from both SSMs are combined using a geometric mean.

$$\Gamma^{(SSM)}(m) = \sqrt{\Gamma^{(SSM, RMS)}(m) \Gamma^{(SSM, MFCC)}(m)}$$

This merged novelty curve peaks at moments of strong musical shift. The algorithm aligns peak positions to the nearest downbeat positions that lie at a multiple of 8 measures from each other because musical phrases are assumed to be 8 measures long. This results in a set of section boundaries.

Track Selection:

The goal of track selection is to determine which songs should be mixed together and in what order, following an intentional progression similar to a human DJ set. There should be a progression from lower-energy tracks, gradually building to a climax, and then easing the listener back down. The algorithm maps each track into a style space derived from spectral and timbral features such as spectral energy and contrast. Songs close together in this space sound compatible, while distance introduces variation.

Using this representation, the system selects songs based on their proximity in the style space. It begins with a random first track, then chooses subsequent songs by estimating an ideal “next style”

as a weighted combination of the current song, the previous song, and the overall style centroid. A small negative weight on the previous song encourages dissimilarity and progression across the mix, allowing it to evolve in sound.

Cue Point Selection:

Cue point selection determines where transitions should occur within each song. These points are chosen to line up with musical sections - the transitions into new sections detected during structural segmentation.

When two songs are mixed, they overlap briefly so that both can be heard, and the system selects entry and exit cue points that maximize rhythmic compatibility. Depending on the context, different transition types can be used:

1. Double Drop: Songs aligned by their drops for maximum impact.
2. Rolling Transition: The buildup of the next song overlaps with the high-energy section of the current one, maintaining flow and energy.
3. Relaxed Transition: A calmer next song starts to transition from an intense section of the current song, giving listeners a break.

A finite state machine chooses which transition type is selected based on the previous mix, ensuring balance. Once a transition type is chosen, cue points are determined for candidate songs, and the system evaluates the quality of the crossfades to select the best pairing for the next mix.

Mixing:

Once the songs and cue points are chosen, the system performs the actual audio mixing. First, beat matching aligns the tempo and phase of both tracks. When tempos differ, time stretching adjusts tempo without altering pitch - fixed tempo of 175 BPM for entire Drum & Bass mix for this implementation.

During the overlap, crossfading gradually decreases the current track's volume while raising the next one's. Concurrently, equalization (EQ) adjusts the balance and relative gain between frequency bands to prevent interference and maintain a clean sound - for example, reducing bass on one track while emphasizing it on the other during overlap.

Together, these steps maintain the clarity and rhythmic alignment characteristic of a human DJ mix.

Summary:

In summary, the automatic DJ algorithm integrates MIR-based beat detection, spectral and feature analysis for segmentation, machine learning for downbeat detection, and style-space navigation for musical sequencing. The resulting system successfully implements an automated Drum and Bass DJ mix.

III. PLAN FOR TESTING AND VALIDATION

There are 3 primary areas of testing:

- Correctly analyzing songs including tempo, structure, key, and cue points
- Selecting compatible tracks that can transition naturally.
- Adding mixes seamlessly without any audible glitches, rhythmic error or mismatched keys.

Demonstration of Functionality:

The system functionality can be demonstrated by feeding it a playlist of songs. This can range from 20-50 electronic dance music songs. Based on these songs and the preprocessing, it should select and mix songs to finally output a 5-10 minute long continuous mix.

Inputs	Description
Audio Files	EDM tracks used for training and testing the DJ system
Musical Data	Label for tempo, downbeats, and section boundaries
Parameter Settings	Crossfade duration, tempo limits

Outputs	Objective/Subjective	Description
Annotated song metadata	Objective	Tempo, beat positions, downbeats, structure segments, keys
Mixed Audio File	Subjective	Seamless continuous audio
Performance Logs	Objective	Processing times, alignment

		errors, beat detection accuracy
User Ratings	Subjective	Listener feedback on quality and naturalness of transitions

Evaluation Metrics:

Component	Description
Beat Tracking	Beat tracking accuracy using F-Measure
Downbeat Detection	Checks if the downbeats align
Tempo Estimation	Difference between detected tempo and actual tempo (within 1 BPM)
Key Detection	Correctly identified keys in the song
Structural Segmentation	Time difference between predicted and actual section boundaries
Transition Timing	How well the two songs' beats align during crossfade

Testing:

The first aspect is to test each individual module to see if they work correctly before integrating them all together. Once we combine the multiple modules, the integrated testing module has to be able to take in a library of songs, and the verification would involve checking the beats and downbeats are consistent across the songs, the structural segmentation aligns with the musical cues, and cue points are chosen at appropriate times. These can be verified using the performance logs. Finally, the system testing will involve analyzing the audio file to ensure that there is beat alignment and tempo stability. We also plan on conducting subjective listening tests where individuals are provided both the system's automated DJ mix and a human DJ mix and listeners will be asked to rate smoothness, energy progression and overall preference as part of the user rating output.

IV. CONCLUSION

All in all, this automated DJ system represents a fusion of music, machine learning, and digital signal processing to replicate the creativity of a human DJ. Through rigorous analysis of the musical structure, we hope that the system can autonomously generate coherent and fluid mixes. In essence, this project not only automates the mechanical tasks of DJing, but also captures the creative essence of the DJing craft. Its applications extend beyond entertainment and provide amateurs with an opportunity to create their own unique musical mixes.

V. REFERENCES

Vande Veire, Len, and Tijl De Bie. "From raw audio to a seamless mix: creating an automated DJ system for Drum and Bass." *EURASIP Journal on Audio, Speech, and Music Processing* 2018.1 (2018): 13.